

Artificial Intelligence Handbook

Python

Python is a high-level, interpreted programming language known for its simplicity and readability. It is widely used in web development, automation, data science, machine learning, and artificial intelligence applications.

Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed.

Artificial Intelligence

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines.

Large Language Models

Large Language Models (LLMs) are AI models trained on massive amounts of text data.

Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with language generation to provide more accurate responses.

LangChain

LangChain is an open-source framework used to build applications powered by large language models.

Vector Databases

Vector databases store numerical representations of text called embeddings.

Embeddings

Embeddings are numerical vector representations of text, images, or other data.

FAISS

FAISS is an open-source library developed by Meta for efficient similarity search of dense vectors.

Groq

Groq is an AI inference platform that provides extremely fast execution of large language models.

Hallucination in AI

Hallucination occurs when a language model generates information that is incorrect or unsupported by context.

Faithfulness

Faithfulness measures whether an AI-generated answer is supported by retrieved context.

Answer Relevancy

Answer relevancy measures how well a generated response addresses the user's question.

Context Precision

Context precision measures how relevant retrieved documents are to the user's query.

RAGAS

RAGAS is an evaluation framework for Retrieval-Augmented Generation systems.

Streamlit

Streamlit is a Python framework for building interactive dashboards and web applications.

Evaluation Pipeline

An evaluation pipeline automatically measures answer quality, retrieval quality, and hallucination rates.